

Efficacy of Venetoclax-Based Therapy in t(11;14)-Positive Relapsed/Refractory Multiple Myeloma: A Systematic Review and Meta-Analysis

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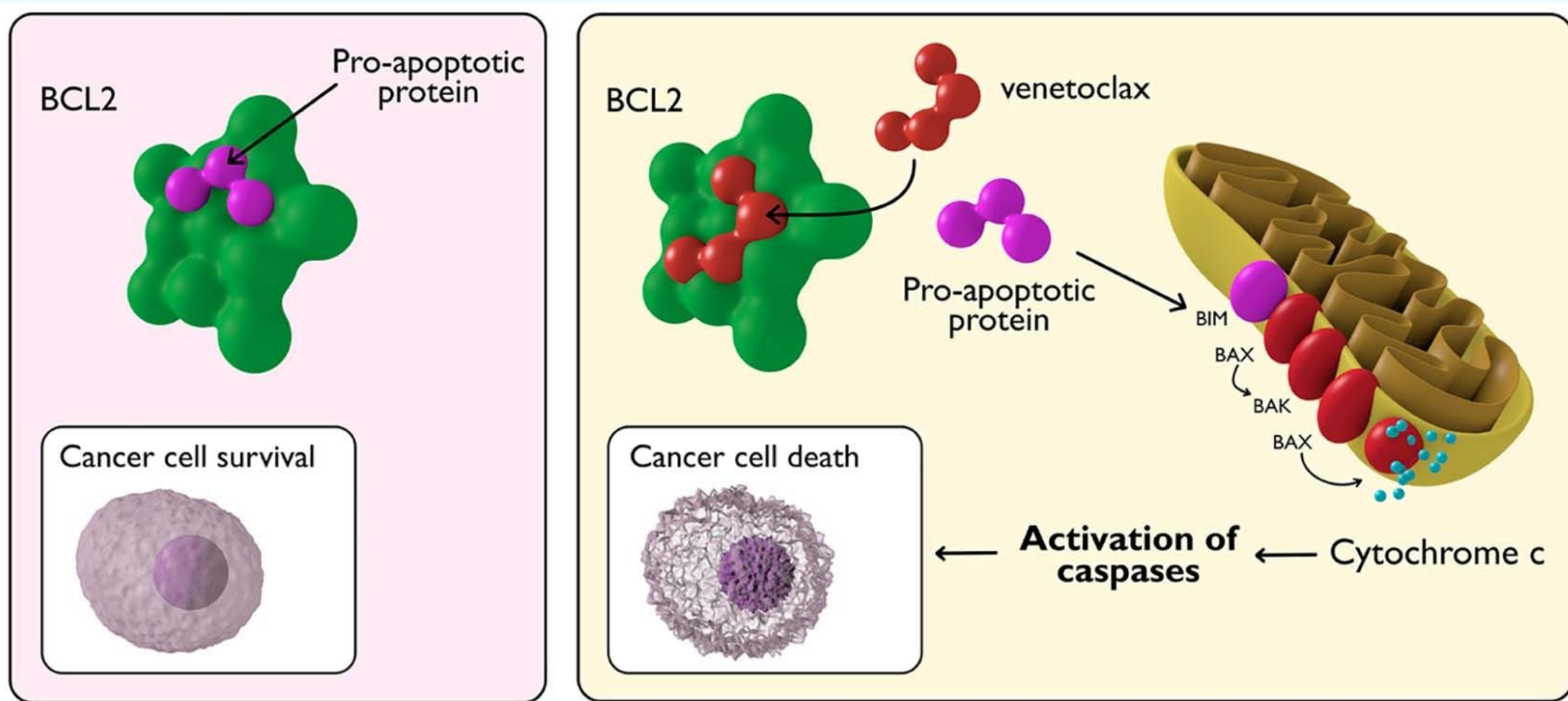
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Purpose

To synthesize randomized phase III evidence evaluating the efficacy of venetoclax-based therapy in patients with t(11;14)-positive relapsed/refractory multiple myeloma (RRMM), with a primary focus on progression-free survival (PFS) and secondary evaluation of overall survival (OS) and time to deterioration in disease-related symptoms (TTDDS).

Background

Translocation t(11;14) is among the most common cytogenetic abnormalities in multiple myeloma and is associated with increased BCL-2 dependence, providing a biologic rationale for targeted inhibition. Venetoclax, a selective BCL-2 inhibitor, has demonstrated promising activity in early-phase studies in this molecularly defined subgroup. However, discordant results from phase III trials and concerns regarding safety have limited regulatory approval. A pooled analysis of randomized data in t(11;14)-positive RRMM is needed to better define its therapeutic role and contextualize trial-level heterogeneity.



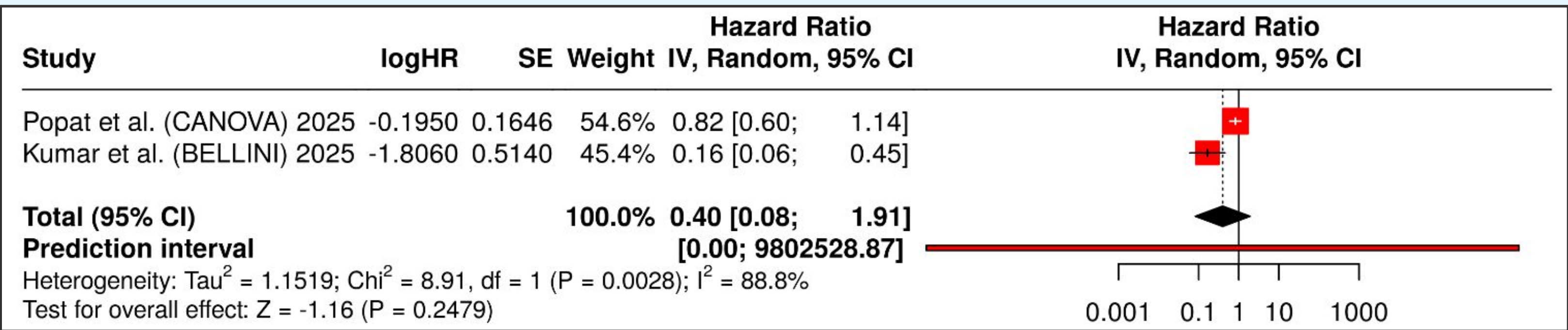
Methods

A systematic literature search was conducted across PubMed-MEDLINE, Embase-OVID, CENTRAL, ClinicalTrials.gov, and Web of Science from inception through January 1, 2025. Randomized phase III trials comparing venetoclax-based regimens with standard-of-care therapies in patients with t(11;14)-positive RRMM were included. Hazard ratios (HRs) for PFS (primary outcome), OS, and TTDDS were extracted and pooled using a random-effects inverse-variance model. Statistical heterogeneity was assessed using the I^2 statistic.

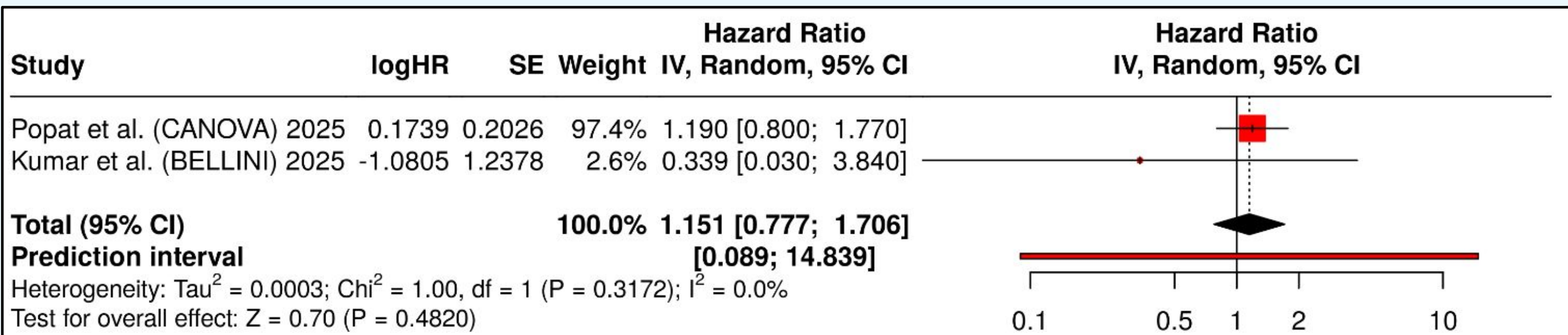
Results

Two phase III randomized controlled trials (CANOVA and BELLINI), comprising 298 patients with t(11;14)-positive RRMM, were included. Venetoclax-based therapy demonstrated a directionally favorable improvement in PFS (HR 0.40; 95% CI, 0.08-1.91), though this did not reach statistical significance and was associated with substantial heterogeneity ($I^2 = 88.8\%$). No significant difference in OS was observed (HR 1.15; 95% CI, 0.78-1.71; $I^2 = 0\%$). A trend toward improved TTDDS was noted (HR 0.70; 95% CI, 0.46-1.05; $I^2 = 0\%$).

Progression-Free Survival (PFS)



Overall Survival (OS)



Time To Deterioration in Disease-Related Symptoms (TTDDS)

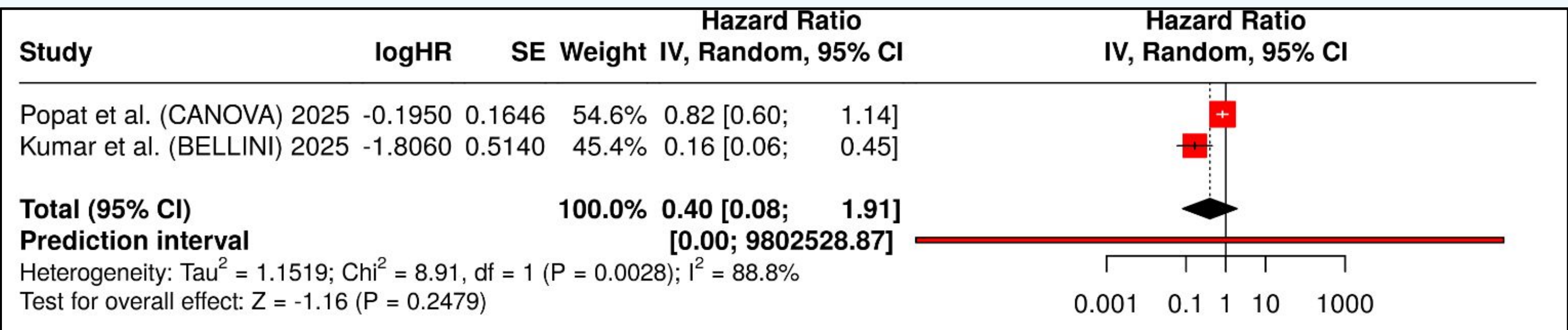


Figure 1 Summary of Pooled Outcomes

Implications

Venetoclax-based therapy demonstrates biologically plausible and directionally favorable efficacy signals in t(11;14)-positive RRMM, particularly for disease control and symptom progression. The absence of statistical significance likely reflects heterogeneity in trial design, comparator regimens, and limited subgroup sample sizes rather than lack of therapeutic activity.

